

**Your grade: 90%**

Your latest: 87.50% • Your highest: 90% • To pass you need at least 80%. We keep your highest score.

Next item →

1. Which of the following do you typically see in a ConvNet? (Check all that apply.)

1 / 1 point

☒ Multiple CONV layers followed by a POOL layer

Nice work

True, as seen in the case studies.

☐ FC layers in the first few layers☒ FC layers in the last few layers

Nice work

True, fully-connected layers are often used after flattening a volume to output a set of classes in classification.

☐ Multiple POOL layers followed by a CONV layer

2. In LeNet - 5 we can see that as we get into deeper networks the number of channels increases while the height and width of the volume decreases. True/False?

1 / 1 point

☒ True☐ False

Nice work

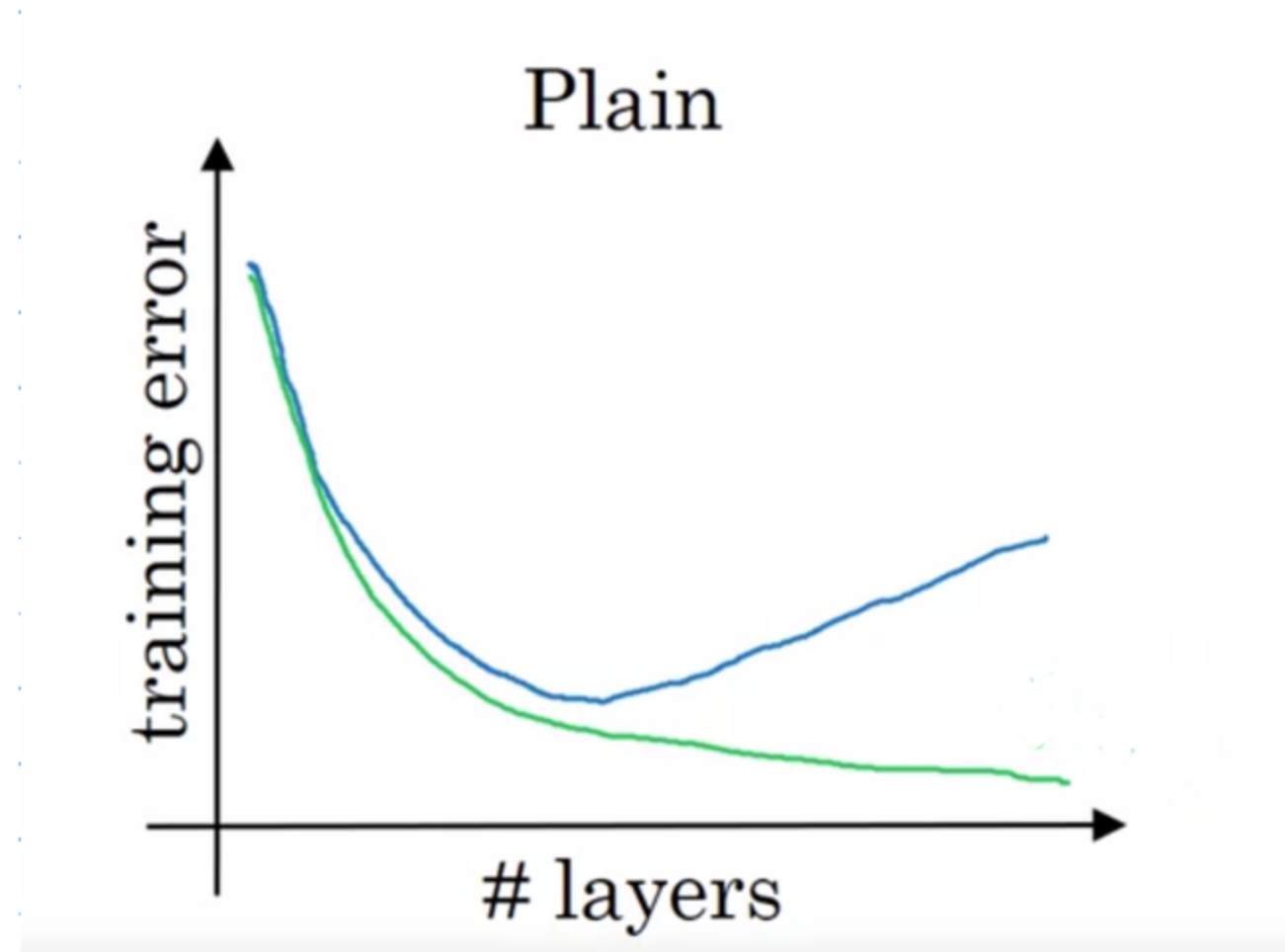
Correct, since in its implementation only valid convolutions were used, without padding, the height and width of the volume were reduced at each convolution. These were also reduced by the POOL layers, whereas the number of channels was increased from 6 to 16.

3. Based on the lectures, in the following picture, which curve corresponds to the expected behavior in theory, and which one corresponds to the behavior we get in practice? This when using plain neural networks.

1 / 1 point

3. Based on the lectures, in the following picture, which curve corresponds to the expected behavior in theory, and which one corresponds to the behavior we get in practice? This when using plain neural networks.

1 / 1 point



- ☐ The blue one depicts the theory, and the green one the reality.
- ☐ The blue one depicts the results in theory, and also in practice.
- ☐ The green one depicts the results in theory, and also in practice.
- ☒ The green one depicts the results in theory, and the blue one the reality.

Nice work

Yes, in theory, we expect that as we increase the number of layers the training error decreases; but in practice after a certain number of layers the error increases.



4. Which of the following equations captures the computations in a ResNet block?

1 / 1 point

- ☒ $a^{[l+2]} = g(W^{[l+2]} g(W^{[l+1]} a^{[l]} + b^{[l+1]}) + b^{[l+2]} + a^{[l]})$
- ☐ $a^{[l+2]} = g(W^{[l+2]} g(W^{[l+1]} a^{[l]} + b^{[l+1]}) + b^{[l+2]} + a^{[l]}) + a^{[l+1]}$
- ☐ $a^{[l+2]} = g(W^{[l+2]} g(W^{[l+1]} a^{[l]} + b^{[l+1]}) + b^{[l+2]}) + a^{[l]}$
- ☐ $a^{[l+2]} = g(W^{[l+2]} g(W^{[l+1]} a^{[l]} + b^{[l+1]}) + b^{[l+2]})$

✔ Nice work

Correct. This expresses the computations of a ResNet block, where the last term $a^{[l]}$ is the shortcut connection.

5. Adding a ResNet block to the end of a network makes it deeper. Which of the following is true?

1 / 1 point

- ☐ It shifts the behavior of the network to be more like the identity function.
- ☐ The performance of the networks is hurt since we make the network harder to train.
- ☐ The number of parameters will decrease due to the shortcut connections.
- ☒ The performance of the networks doesn't get hurt since the ResNet block can easily approximate the identity function.

✔ Nice work

Yes, as noted in the lectures in a ResNet block the computations are given by $a^{[l+2]} = g(W^{[l+2]} a^{[l+1]} + b^{[l+2]} + a^{[l]})$ thus if $W^{[l+2]}$ and $b^{[l+2]}$ are zero then we get the identity function.

6. 1×1 convolutions are the same as multiplying by a single number. True/False?

1 / 1 point

- ☐ True
- ☒ False

✔ Nice work

Yes, a 1×1 layer doesn't act as a single number because it makes a sum over the depth of the volume



7. Which ones of the following statements on Inception Networks are true? (Check all that apply.)

0.8 / 1 point

- ☒ Making an inception network deeper (by stacking more inception blocks together) can improve performance, but can also lead to overfitting and increase in computational cost.

✔ Nice work

- ☐ Inception networks incorporate a variety of network architectures (similar to dropout, which randomly chooses a network architecture on each step) and thus has a similar regularizing effect as dropout.

- ☒ A single inception block allows the network to use a combination of 1x1, 3x3, 5x5 convolutions and pooling.

✔ Nice work

- ☐ Inception blocks usually use 1x1 convolutions to reduce the input data volume's size before applying 3x3 and 5x5 convolutions.

✘ You didn't select all the correct answers

8. When having a small training set to construct a classification model, which of the following is a strategy of transfer learning that you would use to build the model?

1 / 1 point

- ☒ Use an open-source network trained in a larger dataset freezing the layers and re-train the softmax layer.
- ☐ Use an open-source network trained in a larger dataset, freeze the softmax layer, and re-train the rest of the layers.
- ☐ Use an open-source network trained in a larger dataset. Use these weights as an initial point for the training of the whole network.
- ☐ It is always better to train a network from a random initialization to prevent bias in our model.

✔ Nice work

Yes, this is a strategy that can provide a good result with small data.

✔ Nice work

Yes, this is a strategy that can provide a good result with small data.

9. Which of the following are true about Depthwise-separable convolutions? (Choose all that apply)

0 / 1 point

☒ The pointwise convolution convolves the output volume with 1×1 filters.

✔ Nice work

Yes, the number of filters for the output of the depthwise-separable convolution is determined by the number of 1×1 filters used.☐ The depthwise convolution convolves each channel in the input volume with a separate filter.☐ The depthwise convolution convolves the input volume with 1×1 filters over the depth dimension.☒ Depthwise-separable convolutions are composed of two different types of convolutions.

✔ Nice work

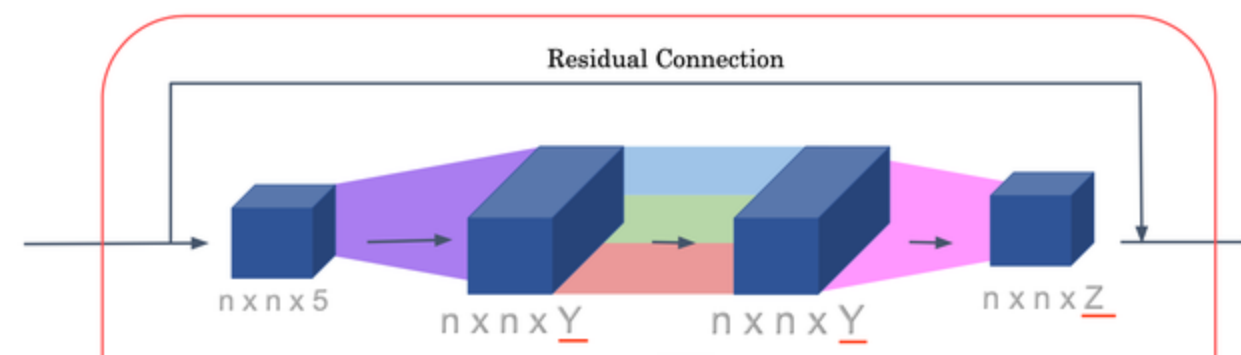
Yes, it is composed of a depthwise convolution followed by a pointwise convolution.

✘ You didn't select all the correct answers

10. Fill in the missing dimensions shown in the image below (marked W, Y, Z).

1 / 1 point

MobileNet v2 Bottleneck



✓ Nice work

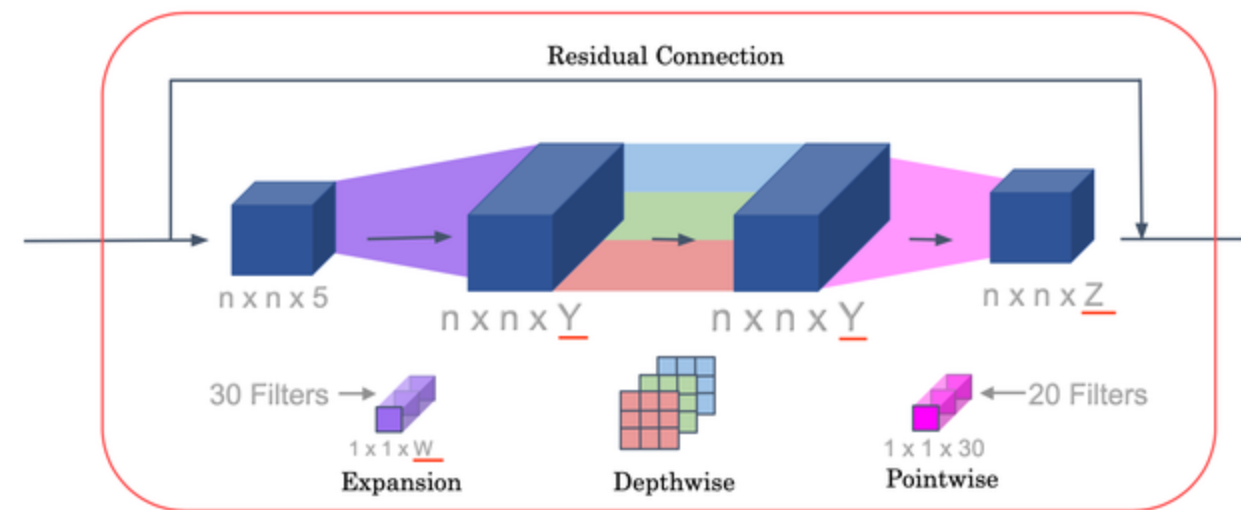
Yes, it is composed of a depthwise convolution followed by a pointwise convolution.

⊗ You didn't select all the correct answers

10. Fill in the missing dimensions shown in the image below (marked W, Y, Z).

1 / 1 point

MobileNet v2 Bottleneck



☐ W = 30, Y = 20, Z = 20

☐ W = 5, Y = 20, Z = 5

☒ W = 5, Y = 30, Z = 20

☐ W = 30, Y = 30, Z = 5

✓ Nice work